Reference Pricing as a Focal Point: Evidence from the Finnish Pharmaceutical Market

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Introduction

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- Firms adapt to regulation.

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- The regulator needs to adjust regulation in response.



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  - Intended: Keep prices low
  - Unintended: Observable **focal point** for coordination (Knittel and Stango, 2003; Genakos et al., 2018; Martin and Verboven, 2026)

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- Finnish pharmaceutical reference price system
  - Maximum price cap – set by the regulator, fixed, similar to price ceiling
  - **Reference price** – set endogenously by firms' prices, moving

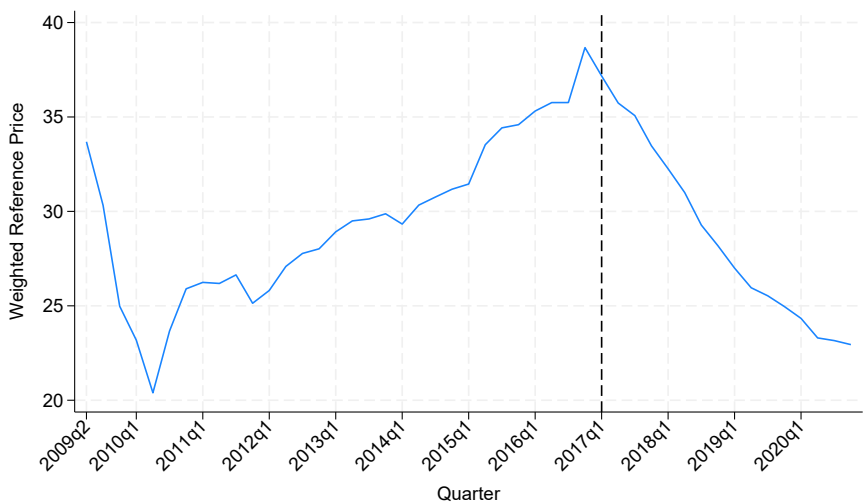
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- This paper
  - Reference Pricing as a Focal Point
  - Tug-of-war between firms and the regulator

Introduction

- The system was introduced in 2009. Average RP dropped.

Figure: Average Reference Price

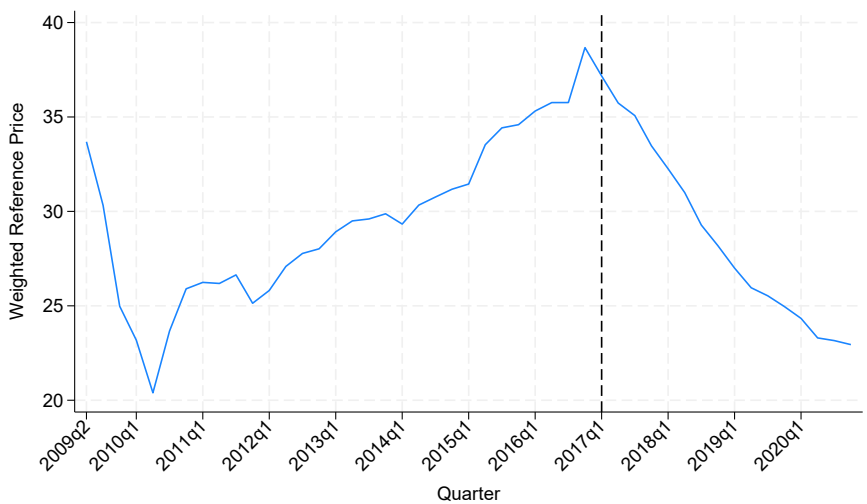


Notes: Average reference price weighted by sales in 2016Q4. Data source: AFP (2009—2022), Hila (2009–2022), Fimea (2009–2020).

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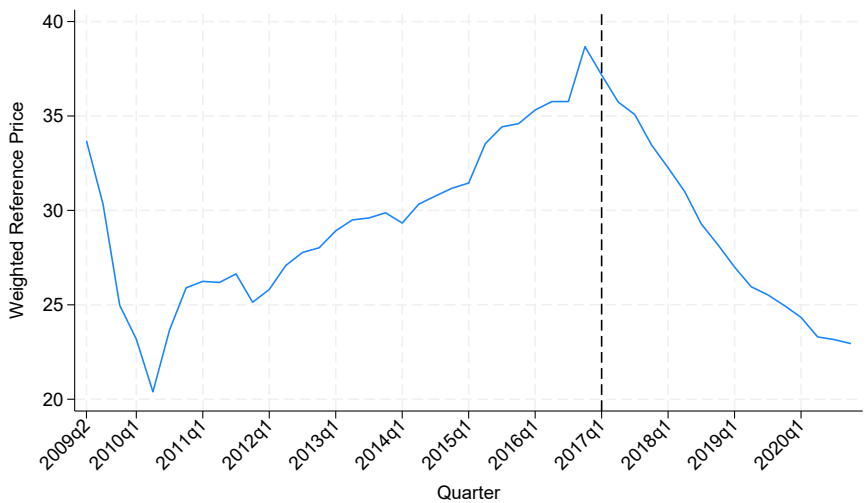


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- In 2017, the regulator made a minor change to the system. Why was it effective?

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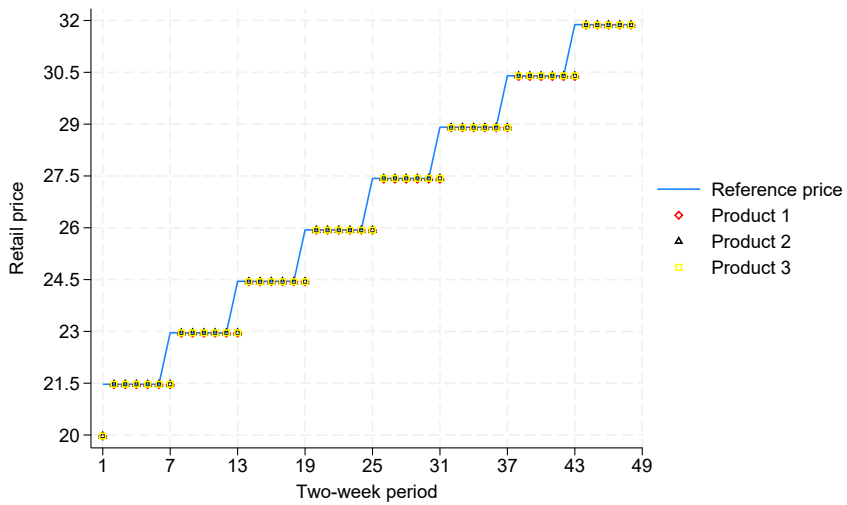
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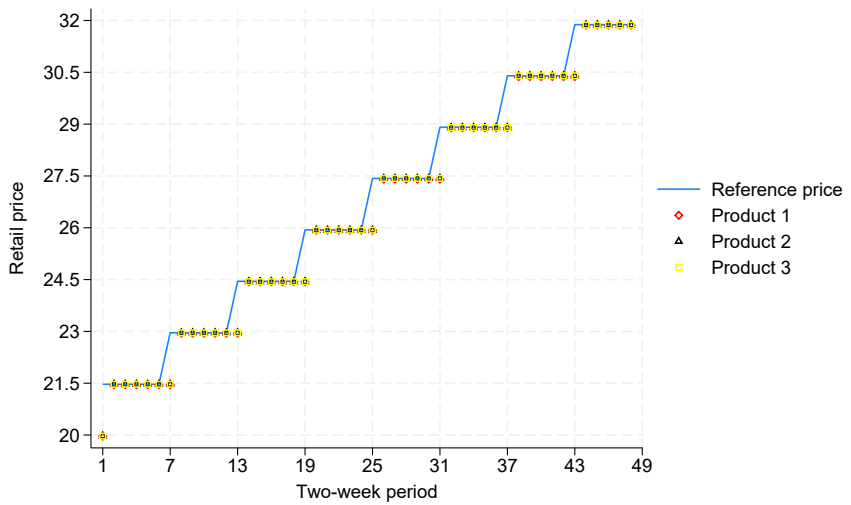


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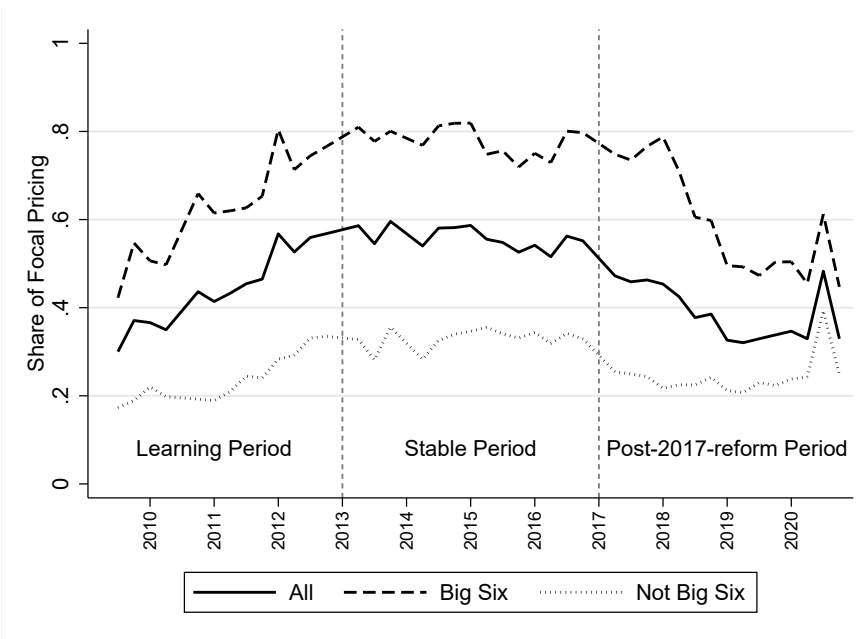
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Figure: Share of Products using Focal-point Pricing



Notes: Data source: Fimea (2009–2020).

Results Summary

- Reference Price (RP)
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After the RP is fixed	7% priced above the RP; 50% priced exactly at the RP.	

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After the RP is fixed	7% priced above the RP; 50% priced exactly at the RP.	The RP constrained 83% of prices at the RP.
2017 regulation	$x = 1.5/2 \rightarrow 0.5$.	Easier to push rivals above the RP.

Outline

1. Related Literature
2. Regulations & Data
3. The Big Six
4. Descriptive Analyses
5. Structural Model
6. Conclusions

Literature

Related Literature

- Focal-point pricing through regulation: Knittel and Stango (2003); Genakos et al. (2018); Martin and Verboven (2026).

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- Reference price regulation: Brekke et al. (2009, 2011); Kaiser et al. (2014); Granlund and Bergman (2018); Kortelainen et al. (2023).

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- Pricing patterns and collusion in pharmaceutical markets: Hauschultz and Munk-Nielsen (2020); Janssen (2022); Clark et al. (2022); Starc and Wollmann (2025); Le and Toivanen (2026).

  Our paper: the regulator adapts to firms' pricing behavior.

Regulations

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- Maximum price caps: Set by the regulator.

Regulations - Reference Price

- If a product's price exceeds the reference price:
  - Generic substitution: Pharmacies offer substitution to a cheaper alternative.
  - Reference pricing (from 4/2009): Consumers pay the exceeding part above the reference price fully out-of-pocket.

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- Example reference price group with a price band of €1.5 and 60% copayment patient:

Product	Retail price	Reference price	Out-of-pocket cost
A	€15	€10	$10 \times 0.6 + (15 - 10) = €11$
B	€10	€10	$10 \times 0.6 = €6$
C	€8.5	€10	$8.5 \times 0.6 = €5.1$

Data

- Product-level wholesale monthly data
- Finland (AFP, Fimea) and Norway (Farmastat) (to construct demand instruments)
- Prices, sales, product characteristics
- 1 483 markets from 4/2009 to 1/2022.

Big Six

The Big Six

- Biggest generic firms in the market.
- Prices equal to the previous quarter's reference price at the beginning of the quarter (focal-point pricing).

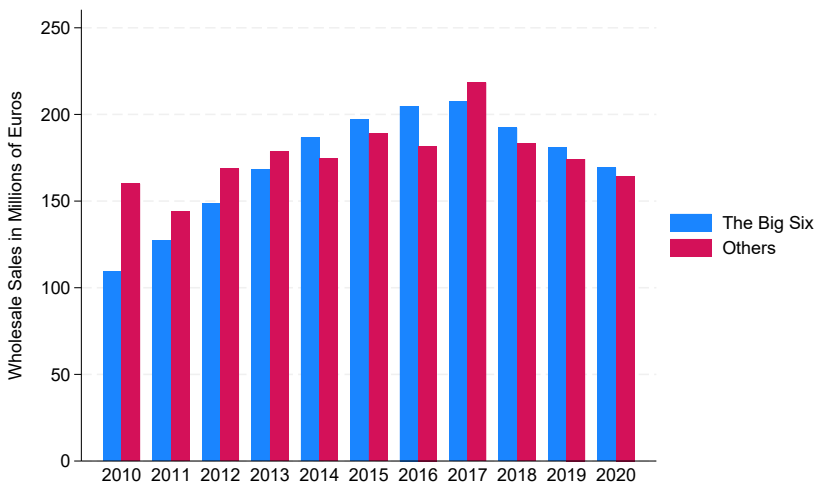
Sales

- The Big Six is an important part of the Finnish reference price system.

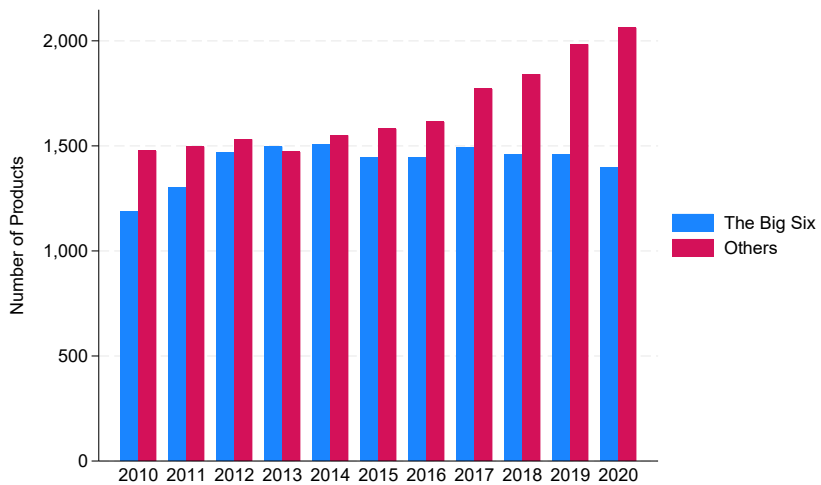
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a) Sales in Millions of Euros



b) Number of Unique Products



Data Source: AFP (2010–2020), Fimea (2010–2020).

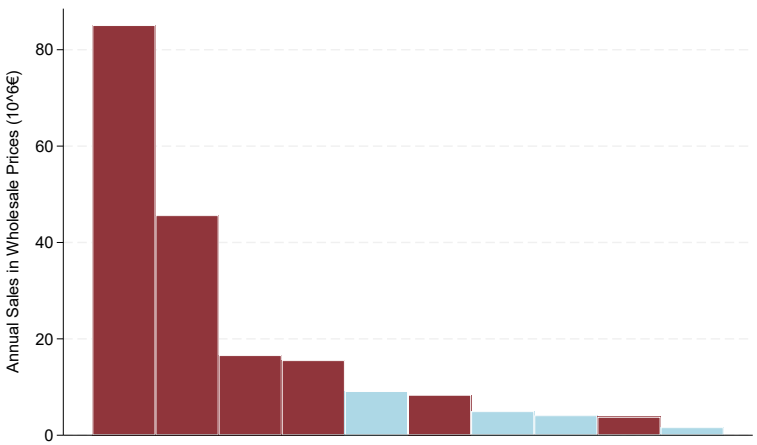
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- We rank the ten biggest firms in 2013 by sales and by number of products.

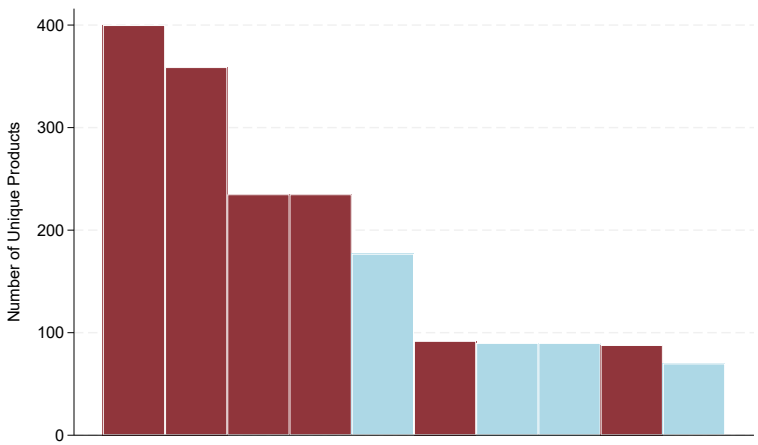
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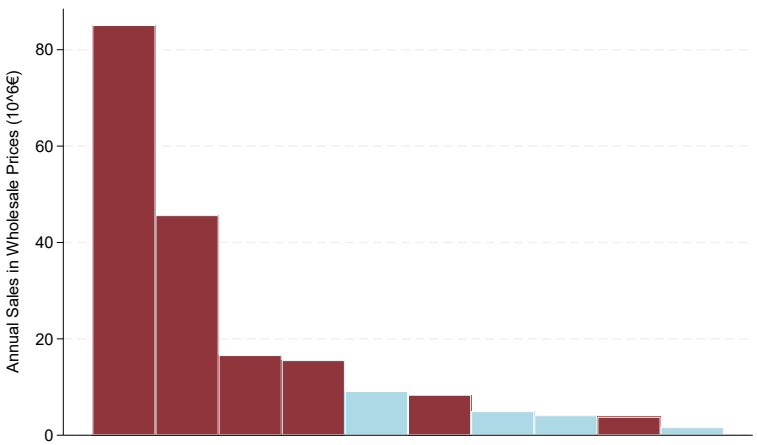


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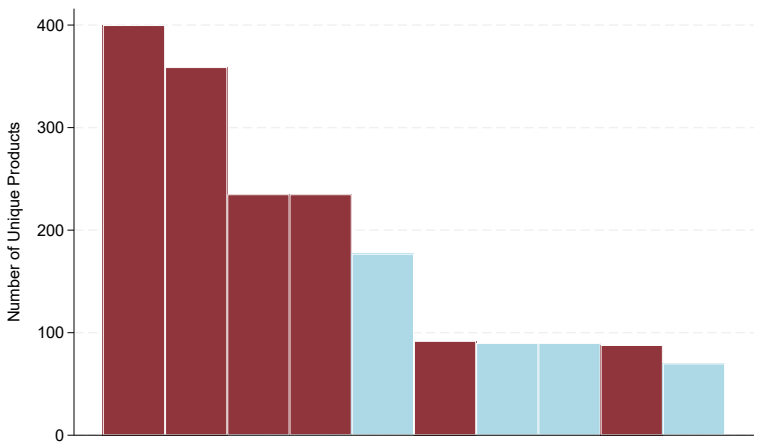
Sales

- We rank the ten biggest firms in 2013 by sales and by number of products.
- The two smallest firms in the Big Six were merged with two of the biggest four.

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4. **Descriptive Analyses**
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Descriptive Analyses

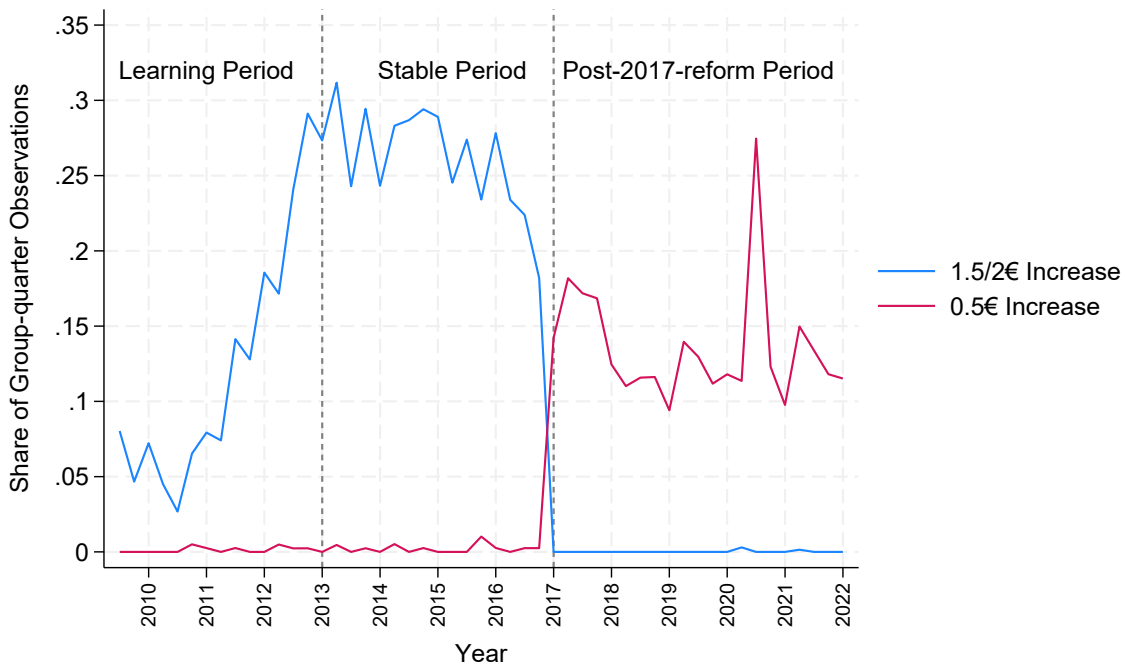
Focal-point Coordination

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- Before 2017: €1.5/2 increases were common.
- After 2017: €0.5 increases were common.

Figure: Share of Markets with Focal-point Coordination



Notes: Data source: AFP (2009–2022).

Market Type

- Type 1: Both types of firms
- Type 2: Only the other firms
- Type 3: Only Big Six firms

Descriptive Regressions of Reference-price Increases

$$RPIncrease_{mt} = \alpha_0 + \Sigma_{j=1}^{3} \alpha_j Typej_{mt} + \Sigma_{j=1}^{3} \beta_j RPRatio_{mt} \times Typej_{mt} + \gamma Big6Ratio_{mt} \times Type1_{mt} + \delta_0 Market_m + \delta_1 Time_t \quad (1)$$

- LHS: dummy of having an increase in RP

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  - RPRatio: Last Reference Price / Price Cap.
  - Big6Ratio: Share of products from the Big Six in the market (only relevant for Market Type 1).

Descriptive Regressions of Reference-price Increases

	(1)
	RP Increase
Constant	.23
Panel 1: Market Types	
Type 2 (Only the other firms)	.02
	(.02)
Type 3 (Only Big Six)	.29***
	(.02)
Market and Time FEs	Yes
Observations	17654

Notes: Robust SEs in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Data source: Fimea (2009–2020).

Descriptive Regressions of Reference-price Increases

	(1) RP Increase	(2) RP Increase
Constant	.23	.49
Panel 1: Market Types		
Type 2 (Only the other firms)	.02	.12***
	(.02)	(.04)
Type 3 (Only Big Six)	.29***	.57***
	(.02)	(.05)
Panel 2: Reference Price / Price Cap Ratios		
Reference Price Ratio × Type 1 (Mix of firms)		-.38***
		(.02)
Reference Price Ratio × Type 2 (Only the other firms)		-.50***
		(.05)
Reference Price Ratio × Type 3 (Only Big Six)		-.74***
		(.07)
Market and Time FEs	Yes	Yes
Observations	17654	17654

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Descriptive Regressions of Reference-price Increases

	(1) RP Increase	(2) RP Increase	(3) RP Increase
Constant	.23	.49	.28
Panel 1: Market Types			
Type 2 (Only the other firms)	.02	.12***	.27***
	(.02)	(.04)	(.04)
Type 3 (Only Big Six)	.29***	.57***	.83***
	(.02)	(.05)	(.06)
Panel 2: Reference Price / Price Cap Ratios			
Reference Price Ratio × Type 1 (Mix of firms)		-.38***	-.38***
		(.02)	(.02)
Reference Price Ratio × Type 2 (Only the other firms)		-.50***	-.50***
		(.05)	(.05)
Reference Price Ratio × Type 3 (Only Big Six)		-.74***	-.74***
		(.07)	(.07)
Panel 3: Big Six Ratio			
Big Six Ratio × Type 1 (Mix of firms)			.41***
			(.04)
Market and Time FEs	Yes	Yes	Yes
Observations	17654	17654	17654

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Probabilities of Reference Price Increases

Market Types	(1)	(2)	(3)
Type 1 (Mix of firms)	23%	25%	25%
Type 2 (Only the other firms)	25%	21%	15%
Type 3 (Only Big Six)	52%	51%	56%

Notes: Using the average RPRatio and Big6Ratio for each market type. Data source: Fimea (2009–2020).

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Structural Model

Demand Model

- Indirect utility function:

$$v_{ij} = \alpha + \gamma R_j - \beta_1 ln(P_j|R_j = 1) - \beta_2 ln(P_j|R_j = 0) + \kappa X_j + \xi_j + \epsilon_{ij}$$
$$= \delta_j + \epsilon_{ij} \quad (2)$$

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  - Priced at or below the reference price
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- 2 price variables
  - Difference in the reimbursement and pharmacy recommendation status
- X_j includes product characteristics.
- This yields the standard market share expression:

$$s_j = \frac{exp(\delta_j)}{1 + \sum_k exp(\delta_k)} \quad (3)$$

Demand Estimation

- Estimating equation

$$ln(s_{jmt}) - ln(s_{0mt}) = \alpha + \gamma R_{jmt} - \beta_1 ln(P_{jmt}|R_{jmt} = 1) - \beta_2 ln(P_{jmt}|R_{jt} = 0) + \kappa X_{jmt} + \mu_m + \tau_t + \xi_{jmt} \quad (4)$$

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- s_0: 5% outside market share.
- Instruments: matched / imputed price of the same product in Norway, dummy of having a match, number of products in the market.

Demand Estimation

	(1)
	Logit
At or below reference price (0/1)	7.83***
	(.65)
At or below reference price σ	
log(Price) — At or below reference price	-2.01***
	(.23)
log(Price) σ — At or below reference price	
log(Price) — Above reference price	-1.22***
	(.11)
Branded	4.02***
	(.18)
Parallel Import	-.81***
	(.09)
Market and Month FEs	Yes
Observations	80171

Notes: Observations are at the product level. The base category is Generic. Robust SEs in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Data source: Fimea (2009–2020).

Demand Estimation

	(1) Logit	(2) RC Logit	(3) RC Logit
At or below reference price (0/1)	7.83***	9.53***	9.87***
	(.65)	(1.17)	(1.32)
At or below reference price σ		1.93***	
		(.66)	
log(Price) — At or below reference price	-2.01***	-2.14***	-2.31***
	(.23)	(.26)	(.30)
log(Price) σ — At or below reference price			0.65***
			(.24)
log(Price) — Above reference price	-1.22***	-1.30***	-1.33***
	(.11)	(.13)	(.14)
Branded	4.02***	4.67***	4.71***
	(.18)	(.42)	(.44)
Parallel Import	-.81***	-1.57***	-1.67***
	(.09)	(.45)	(.52)
Market and Month FEs	Yes	Yes	Yes
Observations	80171	80171	80171

Notes: Observations are at the product level. The base category is Generic. Robust SEs in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Data source: Fimea (2009–2020).

Marginal Costs

- For each product: minimum observed price over the sample period (Dubois et al., 2022; Tunçel, 2025).

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  - Avoid imposing an assumption on firms' pricing behavior coordinated.

Supply

- Reference-price-setting periods
- Fixed-reference-price periods

Supply

- Reference-price-setting periods: Firms' prices affect the reference price for the whole quarter.
- Fixed-reference-price periods: Firms only face the standard trade-off between markups and market shares in a differentiated product market.

Supply – Reference-price-setting Periods

- Incentives to submit a sufficiently low price to become the only product eligible for full reimbursement and pharmacy recommendation?

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  - Consequence: Overstating the profitability of deviations and, consequently, potential coordination.

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  - Use prices at the end of the previous quarter.
  - Firms were unlikely to change price at the turn of the quarter (13%).

Supply – Reference-price-setting Periods

Table: Observed Deviations and Profitability of Deviations

	Deviation		No Deviation	
	Not Profitable	Profitable	Not Profitable	Profitable
	(1)	(2)	(3)	(4)

Notes: Data source: Fimea (2009–2020).

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Panel 2: Firm Types				
Big 6	0.3%	5.1%	36.4%	**58.2%**
Other firms	1.4%	20.4%	32.9%	**45.3%**

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Panel 3: Market Types				
Type 1 – Big 6	0.3%	5.1%	37.2%	**57.4%**
Type 1 – Non-Big 6	1.4%	18.6%	33.7%	**46.3%**
Type 2 – Non-Big 6	1.6%	28.2%	29.1%	**41.1%**
Type 3 – Big 6	0.5%	5.5%	25.2%	**68.9%**

Notes: Data source: Fimea (2009–2020).

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Panel 4: Time Period				
Learning Period (2009–2012)	1.4%	**9.3%**	53.0%	**36.3%**
Stable Period (2013–2016)	0.8%	**10.9%**	28.1%	**60.3%**
Post-2017-reform Period (2017–2020)	0.6%	**16.7%**	27.4%	**55.3%**

Notes: Data source: Fimea (2009–2020).

Supply – Fixed-reference-price Periods

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Supply – Fixed-reference-price Periods

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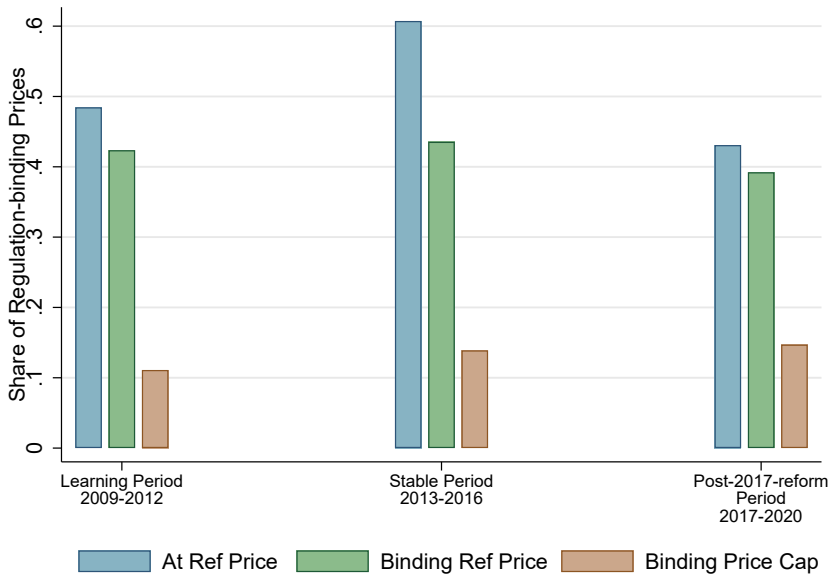
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  - Unilaterally raised their prices above the reference price?
  - Unilaterally raised their prices to the maximum price cap?

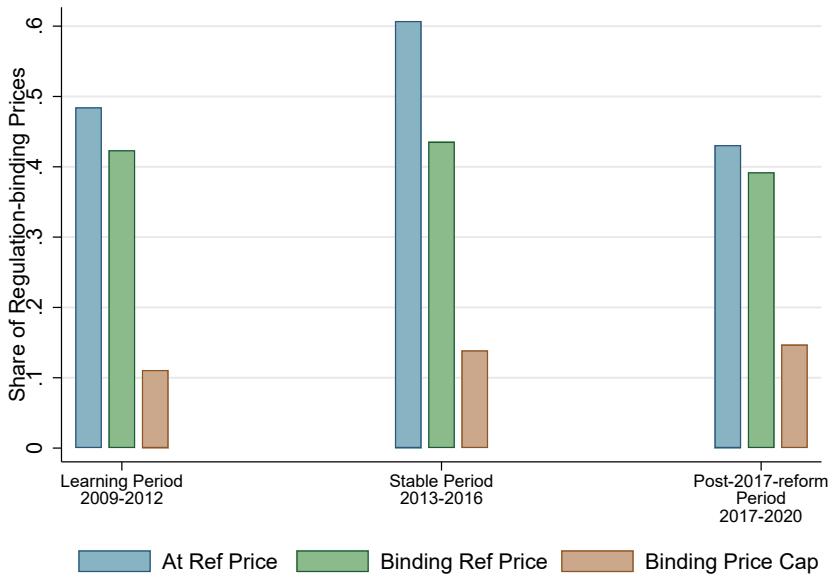
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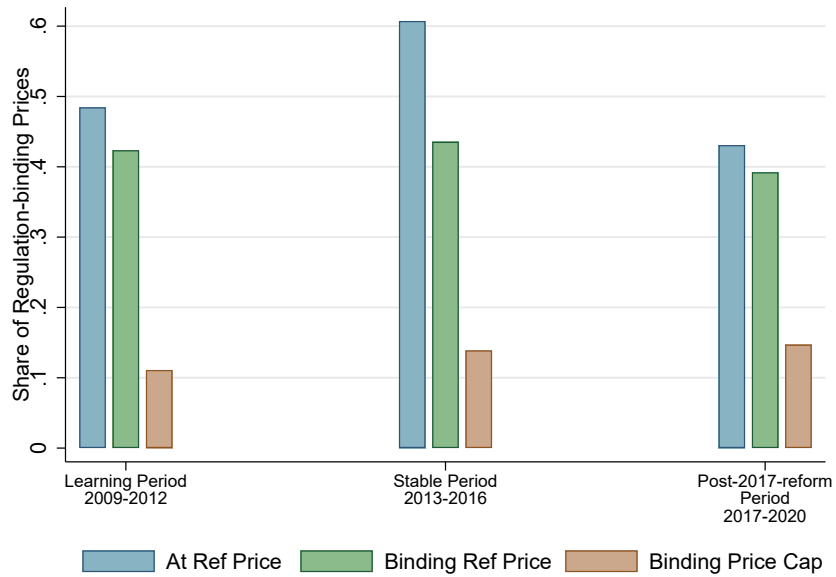
- 40-60% of prices were equal to the reference price.

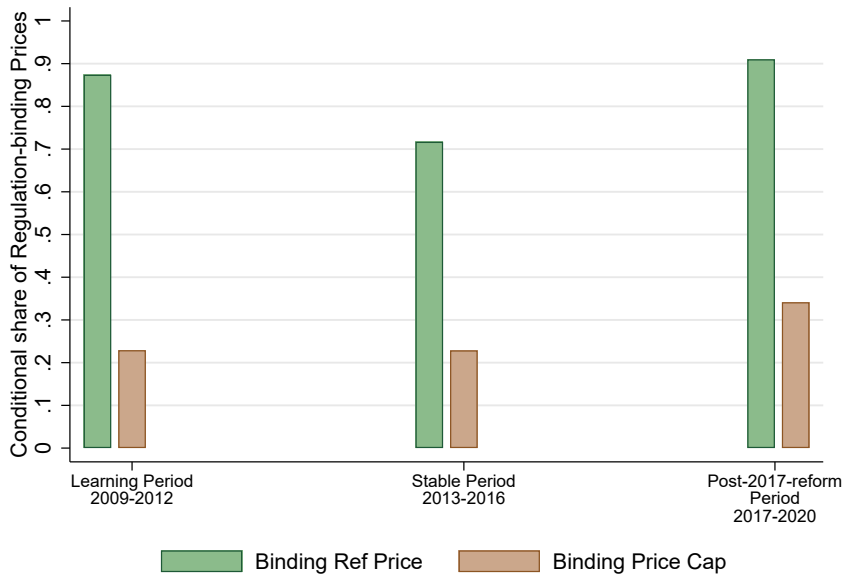


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Supply – Fixed-reference-price Periods

- 40-60% of prices were equal to the reference price.
- Conditional on being priced at the reference price, 71%-91% of them would have priced even higher.

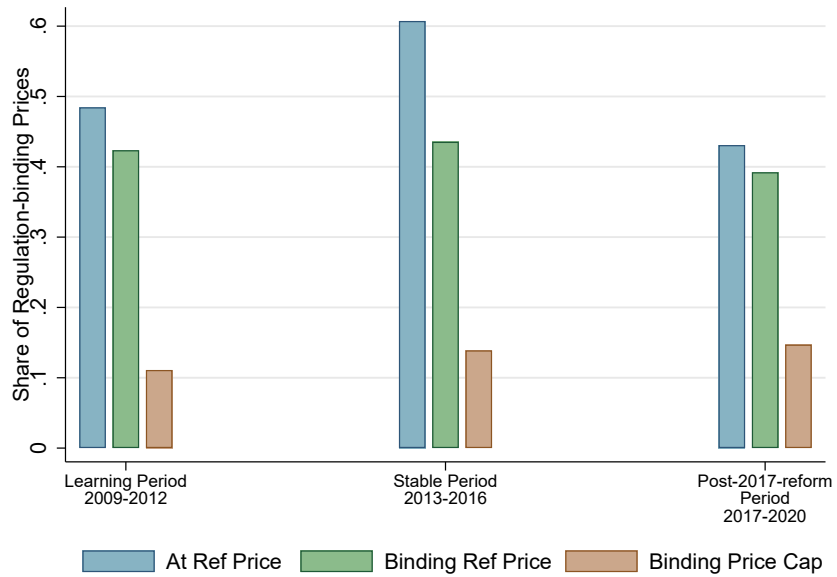


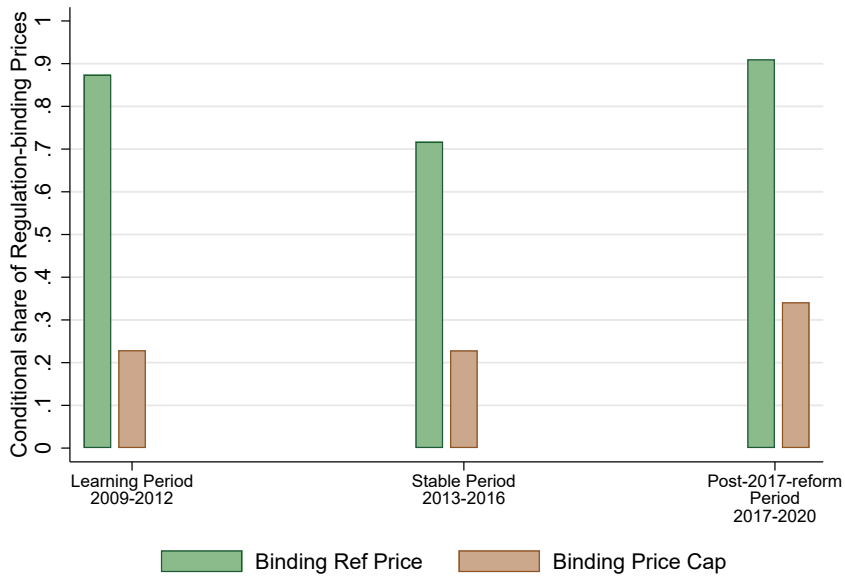


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Supply – Fixed-reference-price Periods

- 40-60% of prices were equal to the reference price.
- Conditional on being priced at the reference price, 71%-91% of them would have priced even higher.
- Without reference pricing, 21%-34% of them would have priced at the maximum price cap.





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Conclusions

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Conclusions

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2. The fixed reference price prevented many firms from increasing prices higher within the quarter.
3. The narrower price band introduced in 2017 increased price competition by reducing firms' incentives to use focal-point pricing.

The regulator needs to constantly respond to firms' pricing behavior.

Brekke, K. R., A. L. Grasdal, and T. H. Holmås: 2009, 'Regulation and pricing of pharmaceuticals: reference pricing or price cap regulation?'. *European Economic Review* **53**(2), 170–185.

Brekke, K. R., T. H. Holmas, and O. R. Straume: 2011, 'Reference pricing, competition, and pharmaceutical expenditures: theory and evidence from a natural experiment'. *Journal of Public Economics* **95**(7-8), 624–638.

Byrne, D. P. and N. De Roos: 2019, 'Learning to coordinate: A study in retail gasoline'. *American Economic Review* **109**(2), 591–619.

Clark, R., C. Fabiilli, and L. Lasio: 2022, 'Collusion in the US generic drug industry'. *International Journal of Industrial Organization* **85**, 102878.

Dubois, P., A. Gandhi, and S. Vasserman: 2022, 'Bargaining and international reference pricing in the pharmaceutical industry'. Technical report, National Bureau of Economic Research.

Genakos, C., P. Koutroumpis, and M. Pagliero: 2018, 'The impact

of maximum markup regulation on prices'. *The Journal of Industrial Economics* **66**(2), 239–300.

Granlund, D. and M. A. Bergman: 2018, 'Price competition in pharmaceuticals–evidence from 1303 Swedish markets'. *Journal of health economics* **61**, 1–12.

Hauschultz, F. P. and A. Munk-Nielsen: 2020, 'The role of demand in price cycles: Evidence from Danish pharmaceutical markets'. *Unpublished Manuscript. https://sites. google. com/view/munk-nielsen/research*.

Janssen, A.: 2022, 'Price dynamics of Swedish pharmaceuticals'. *Quantitative Marketing and Economics* **20**(4), 313–351.

Kaiser, U., S. J. Mendez, T. Rønde, and H. Ullrich: 2014, 'Regulation of pharmaceutical prices: evidence from a reference price reform in Denmark'. *Journal of Health Economics* **36**, 174–187.

Knittel, C. R. and V. Stango: 2003, 'Price ceilings as focal points for tacit collusion: Evidence from credit cards'. *American Economic Review* **93**(5), 1703–1729.

Kortelainen, M., J. Markkanen, M. Siikanen, and O. Toivanen: 2023, *The Effects of Price Regulation on Pharmaceutical Expenditure and Availability*, No. 16. Helsinki GSE Discussion Papers.

Le, H. and O. Toivanen: 2026, 'DP21676 Competition or Collusion? Entry Decisions in the Swedish Pharmaceutical Market'. Technical report, CEPR Press.

Martin, S. and F. Verboven: 2026, 'Price Caps and Coordinated Effects: Evidence from Retail Gasoline'. Technical report, Centre for Economic Policy Research.

Starc, A. and T. G. Wollmann: 2025, 'Does entry remedy collusion? Evidence from the generic prescription drug cartel'. *American Economic Review* **115**(5), 1400–1438.

Tunçel, T.: 2025, 'Should We Prevent Off-Label Drug Prescriptions? Empirical Evidence from France'. *Review of Economic Studies* **92**(3), 2027–2070.